Optimal Sizing of PV System and BESS for Smart Household under Stepwise Power Tariff

Lai ZHOU, Yongjun ZHANG, (*Senior Member IEEE*), Kun LI, Xiaoyu XIE

***Abstract*—The study on the sizing of renewable energy generation systems and energy storage systems together in a household considering different price mechanisms can further promote the development of the home energy management system (HEMS). In this paper, the sizing of photovoltaic (PV) system and battery energy storage system (BESS) in household considering stepwise power tariff (SPT) is investigated, To optimize capacity configuration, a HEMS expressed as a mixed-integer nonlinear programming framework is proposed, considering the uncertainty of PV output, the seasonal load profiles, and the subsidy for PV. That the DICOPT solver in GAMS is called on MATLAB is employed to solve this MINLP problem. Finally, the impact of price mechanisms, uncertainty of PV output, and timing of home load are studied with 6 cases. Results indicate that the cost-saving under SPT is highest among SPT, real-time pricing (RTP), and time-of-use pricing (TOU) tariffs. Besides, the cost-saving in stochastic approach (SA) optimization is greater than that in deterministic approach (DA) optimization.**

***Index Terms*—Photovoltaic system, Battery energy storage system, Smart household, Home energy management system, Stepwise power tariff**

I. Introduction

BUILDINIGS consume around 40% of total end-use energy [1]. Hence, the improvement of energy efficiency in buildings can promote the structural transformation of energy utilization and mitigating the carbon footprint. Renewable energy sources (i.e., rooftop photovoltaic and small-scale wind turbine), energy storage system (i.e., BESS) and smart technologies (i.e., smart meter, wireless communication, intelligent home appliances and demand-side management system) making energy consumption in buildings with green, efficiency, and refinement becomes possible [2-3]. Household as a kind of buildings, its energy management system has received wide attention in recent years, due to the advantages of cost-saving for home, high-efficiency for energy utilization, and low carbon emissions for environment.

Home energy management system is defined as an optimal system monitoring and controlling renewable energy generation, energy storage system, and intelligent home appliances in smart house to decrease electricity cost and improve energy utilization efficiency [4-5]. An effective HEMS can decrease electricity cost by nearly 23.1% and reduce residential peak demand by almost 29.6% [6].

It is crucial to equip with suitable capacity of renewable energy generation and ESS in house for a high-efficiency HEMS. In [7], a convex programming framework was proposed to optimize battery sizing in smart home with BESS and PV system. In [8] and [9], HEMSs were expressed as a mixed-integer nonlinear programming framework to optimize sizing and operation strategy of BESS in smart house. In [10], a mixed-integer linear programing framework was developed for the sizing of additional distributed generation and energy storage system under demand response in smart house, while taking seasonal load and PV generation profiles into consideration. In [11], an integer programming framework was proposed to determine the optimal number of PV modules and batteries for stand-alone PV systems in the residential buildings.

The results of capacity planning of renewable energy generation and energy storage system are affected by electricity prices. TOU and RTP tariffs were used widely in optimal sizing of renewable energy generation and BESS [7-10]. However, many tariffs such as SPT [12-14] were not taken into consideration in capacity planning of renewable energy generation and BESS.

This paper will focus on the optimal sizing of renewable energy generation and BESS in house under SPT. A HEMS expressed as a MINLP framework is proposed to optimize capacity configuration of the PV and BESS in households. Besides, the uncertainty of PV generation, seasonal load profiles, and subsidy of PV are taken into consideration. Then, the DICOPT solver in GAMS is called on MATLAB to find to solve the problem. At the last, the comparative study is carried

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out on 4 cases under different pricing and approaches to verify effectiveness of the proposed HEMS.

II. Mathematical Formulation

The capacity of PV, capacity and rated power of BESS as design variables are optimally determined with the aim of minimum the annual electricity cost in HEMS. The most different is that seasonal PV generation, home load profiles, and subsidy of PV are taken into consideration in the proposed HEMS at the same time in one planning among other studies. The mathematical formulations are given as follows.

A. Objective Function

As the objective function, the annual cost of home is defined in (1). It consists of fixed costs and variable costs. The fixed costs are the sum of the equivalent annual investment cost of PV and BESS. Of which, the capacity subsidy are taken into consideration in PV annual investment cost. The annual variable costs are the sum of variable costs in Q scenarios of PV output. Variable costs for each scenario consist of operation cost of PV and BESS, cost of electricity requested from grid, and income from subsidy and selling electricity to grid by PV generation. The calculation formulation for each cost except C_l^{BUY} is shown in (2).

$$\min C = C_{PV} + C_{BESS} + \sum_{l=1}^{Q}(C_{M,l} + C_l^{BUY} - C_{PV,l}^{SELL} - C_{PV,l}^{IN}) \cdot d_l \tag{1}$$

$$\begin{cases} C_{PV} = (A_{PV} - A_{PV}^{IN}) \cdot P_{PV}^{RATE} \cdot K_{PV} \\ C_{BESS} = (A_{BESS}^{ERATE} \cdot E_{BESS}^{RATE} + A_{BESS}^{PRATE} \cdot P_{BESS}^{RATE}) \cdot K_{BESS} \\ C_{M,l} = A_{PV}^{M} \cdot \sum_{t=1}^{24} P_{PV,l}^{t} + A_{BESS}^{M} \cdot \sum_{t=1}^{24} (P_{ESS,l}^{t,ch} + P_{ESS,l}^{t,dch}) \\ C_{PV,l}^{SELL} = e_{grid}^{sell} \cdot \sum_{t=1}^{24} P_{GRID,l}^{t,SELL} \\ C_{PV,l}^{IN} = e_{allo} \cdot \sum_{t=1}^{24} P_{PV,l}^{t} \end{cases} \tag{2}$$

where, C is annual electricity cost of home, C_{PV} is annual investment cost on PV system, C_{BESS} is annual investment cost on BESS, A_{PV} is Initial investment cost on PV system, A_{PV}^{IN} is capacity subsidy of PV, P_{PV}^{RATE} is rated power of PV system, K_{PV} is coefficient for calculating equivalent annual cost of PV, A_{BESS}^{ERATE} is initial investment cost on capacity of BESS, E_{BESS}^{RATE} is capacity of BESS, A_{BESS}^{PRAT} is initial investment cost on power of BESS, K_{BESS} is coefficient for calculating equivalent annual cost of BESS, $C_{M,l}$ is operation cost of PV and BESS in scenario l, C_l^{BUY} is cost of electricity requested from grid in scenario l, $C_{PV,l}^{SELL}$ is selling electricity to grid in scenario l, $C_{PV,l}^{IN}$ is income from subsidy of PV generation in scenario l, d_l is the number of objects allocated to each scenario, A_{PV}^{M} is operation cost of PV, $P_{PV,l}^{t}$ is output power generated by PV at hour t in scenario l, t is index for hour, t=1,2,…,24, A_{BESS}^{M} is operation cost of BESS, $P_{BESS,l}^{t,ch}$ is charging power of BESS at hour t in scenario l, $P_{BESS,l}^{t,dch}$ is discharging power of BESS at hour t in scenario l, e_{grid}^{sell} is price of power send to grid, e_{allo} is subsidy of PV generation, $P_{GRID,l}^{t,SELL}$ is power send to the grid at hour t in scenario l.

The electricity cost under SPT with n steps is described by Eq. (3), in which judgment needs to be done. In order to decrease the difficulty of solving process, the logic variables are introduced to convert the piecewise linear function to an equation (4) and a series of constraints (5). In (4) and (5), x_1, x_2 … x_n correspond to the electricity consumption under each step respectively. And u_1, u_2 … u_n are 0-1 Boolean variables employed to indicate what step the electricity consumption belongs to. Constraints (5) enforces that the electricity consumption only can be one step in a measurement cycle.

$$C_{IBR}^{BUY} = \begin{cases} P_1 \cdot E_{BUY},\ 0 \le E_{BUY} \le O_1 \\ P_1 \cdot O_1 + P_2 \cdot (E_{BUY} - O_1),\ O_1 < E_{BUY} \le O_2 \\ P_1 \cdot O_1 + P_2 \cdot O_2 + (E_{BUY} - O_2) \cdot P_3,\ O_2 < E_{BUY} \end{cases} \tag{3}$$

$$\begin{aligned} C_{IBR}^{BUY} &= u_1 \cdot P_1 \cdot x_1 + u_2 \cdot (P_1 \cdot O_1 + P_2 \cdot (x_2 - O_1)) \\ &+ u_3 \cdot (P_1 \cdot O_1 + P_2 \cdot O_2 + (x_3 - O_2) \cdot P_3) \end{aligned} \tag{4}$$

$$\begin{cases} E_{BUY} = x_1 + x_2 + x_3 \\ 0 \le x_1 \le O_1 \cdot u_1 \\ O_1 \cdot u_2 \le x_2 \le O_2 \cdot u_2 \\ O_2 \cdot u_3 \le x_3 \le N_B \cdot u_3 \\ u_1 + u_2 + u_3 = 1 \\ u_1, u_2, u_3 \in [0,1] \end{cases} \tag{5}$$

where, C_{SPT}^{BUY} is cost of power requested from grid under SPT, P_1, P_2 … P_n represent price for each step, O_1, O_2 … O_{n-1} represent electricity quantity for each step, E_{BUY} is power purchased from gird in one day.

B. Constraints

The constraints of the lower programming include operation constraints of BESS and power balance constraints between home and grid. They are given as follows:

(1) BESS operational constraints

The operation of BESS is constrained by its rated power of inverter and state-of-energy. Besides, BESS can only operate one of states at a time interval, including charging power, discharging power, and not working [15-16]. Hence, binary variables $x_{bess}^{t,ch}$ and $x_{bess}^{t,dch}$ are employed to indicate what state BESS operates. The power limitations for charging and discharging are expressed as (6) and (7). Constraints (8) and (9) avoid BESS charging and discharging simultaneously.

$$0 \le P_{BESS}^{t,ch} \le P_{BESS}^{RATE} \cdot x_{bess}^{t,ch} \tag{6}$$

$$0 \le P_{BESS}^{t,dch} \le P_{BESS}^{RATE} \cdot x_{bess}^{t,dch} \tag{7}$$

$$x_{bess}^{t,ch} + x_{bess}^{t,dch} = 1 \tag{8}$$

$$x_{bess}^{t,ch}, x_{bess}^{t,dch} \in [0,1] \tag{9}$$

where, $P_{BESS}^{t,ch}$ is charging power of BESS at hour t, $P_{BESS}^{t,dch}$ is discharging power of BESS at hour t.

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To ensure the life time of BESS, the state-of-energy of BESS is limited in the 0.2~0.8 of rated capacity of BESS [17-18], as described in (10). SOC^{t}_{BESS} is the state-of-energy of BESS at t expressed as (11) and (12). Eq. (13) enforces that the initial state-of-energy equal to the final state-of-energy for BESS in an operation cycle.

$$\begin{cases} SOC^{\min}_{BESS} \le SOC^{t}_{BESS} \le SOC^{\max}_{BESS} \\ SOC^{\min}_{BESS} = 0.2 \cdot E^{rate}_{BESS} \\ SOC^{\max}_{BESS} = 0.8 \cdot E^{rate}_{BESS} \end{cases} \tag{10}$$

$$SOC^{t+1}_{BESS} = SOC^{t}_{BESS} + P^{t+1,ch}_{BESS} \cdot \eta_{ch} - \frac{P^{t+1,dch}_{BESS}}{\eta_{dch}} \tag{11}$$

$$SOC^{1}_{BESS} = SOC^{\min}_{BESS} + P^{1,ch}_{BESS} \cdot \eta_{ch} - \frac{P^{1,dch}_{BESS}}{\eta_{dch}} \tag{12}$$

$$SOC^{24}_{BESS} = SOC^{1}_{BESS} \tag{13}$$

where, SOC^{min}_{BESS} is minimum state of energy of BESS, SOC^{max}_{BESS} is maximum state of energy of BESS, η_{ch} is charging efficiency of BESS, η_{dch} is discharging efficiency of BESS.

(2) Power balance constraints

The electricity balance equation for HEMS is described in (14). Eq. (14) forces that the power exchanged with the grid is equal to the sum of home load demand, PV output, and power of BESS at each hour in HEMS. It should be mentioned that HEMS can only operate one of states at a time interval, including drawing power from grid, injecting power into grid, and disconnecting with grid [19]. It is ensured by Constraints (15) and (16).

$$\begin{cases} P^{t,BUY}_{GRID} \cdot x^{t,BUY}_{GRID} - P^{t,SELL}_{GRID} \cdot x^{t,SELL}_{GRID} \\ \quad = P^{t}_{load} - P^{t}_{PV} + P^{t,ch}_{BESS} - P^{t,dch}_{BESS} \\ P^{t}_{PV} = P^{t,p.u}_{PV} \cdot P^{RATE}_{PV} \end{cases} \tag{14}$$

$$x^{t,BUY}_{GRID} + x^{t,SELL}_{GRID} = 1 \tag{15}$$

$$x^{t,BUY}_{GRID}, x^{t,SELL}_{GRID} \in [0,1] \tag{16}$$

where, $P^{t,BUY}_{GRID}$ is power purchased from grid at hour t, $P^{t,SELL}_{GRID}$ is power send to grid at hour t, $P^{t,p.u}_{PV}$ is per unit output of PV at hour t, P^{t}_{load} is home load demand at hour t, $x^{t,BUY}_{GRID}$ and $x^{t,SELL}_{GRID}$ is binary variables.

C. Solving the Problem

Due to the nonlinear equation in (3), the proposed HEMS formulation under STP tariff is a MINLP framework. To solve this problem, the DICOPT solver in GAMS is called on MATLAB. The generation of scenarios of PV output and individuals as solution set, calculation of objective function, and selection of best individual are carried out in MATLAB. In calculation of objective function, the minimum variable cost is determined by optimal charging-discharging pattern of BESS in GAMS with DICOPT solver.

III. Test and Results

A. Scenario Generation for PV Output

According to weather conditions and seasons, the annual PV output data are divided into 12 scenarios as follows: Spring sunny, Spring cloudy, Spring rainy, Summer sunny, Summer cloudy, Summer rainy, Autumn sunny, Autumn cloudy, Autumn rainy, Winter sunny, Winter cloudy, and Winter rainy, corresponding to scenario1~scenario 12. Fig.1 presents the PV generation for each scenario.

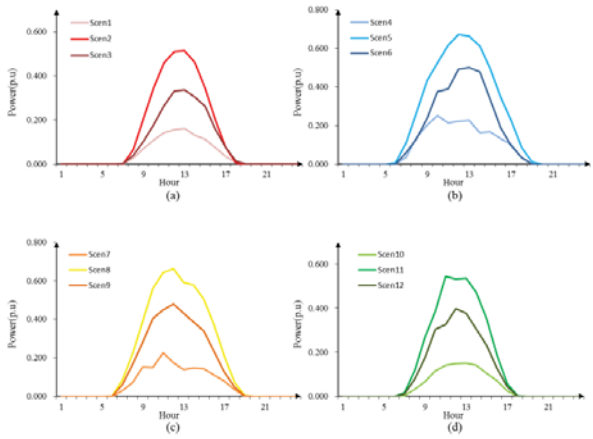

Figure 1 PV generation scenarios in different season

B. Test System

In order to demonstrate the impact of electricity tariffs and seasonal PV generation and load profiles, the comparative study is carried out on 4 cases.

Case 1: optimization under SPT with SA.
Case 2: optimization under RTP with SA
Case 3: optimization under TOU with SA.
Case 4: optimization under SPT with DA.

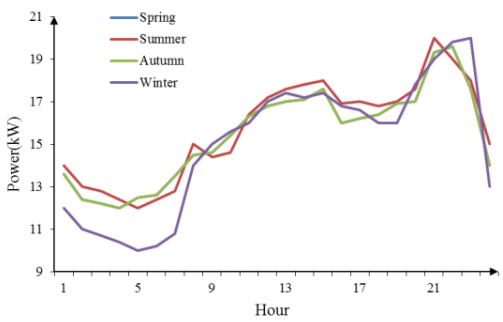

Figure 2 Load profile of home at typical day in spring, summer, autumn, and winter

For SA, the load profile of home at typical day in spring, summer, autumn, and winter is shown in Fig.2. 12 scenarios of PV output power are given in Fig.1 and the weight of each scenario is listed in Tab.1. For DA, the load and output power of PV are depicted in Fig.4. Besides, the SPT in Guangzhou is used in this paper including 3 steps, as shown as Fig.5. It should be mentioned that the monthly electricity steps are converted

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into daily electricity steps. Due to day is schedule cycle of operation optimization of HEMS. Fig. 6 shows the prices under the RTP and TOU tariffs.

TABLE I
THE POPULATION AND WEIGHT OF EACH SCENARIO

Cluster	Numbers	Weight	Cluster	Numbers	Weight
Scen1	44	0.12	Scen7	17	0.046
Scen2	18	0.049	Scen8	36	0.097
Scen3	32	0.089	Scen9	41	0.112
Scen4	27	0.074	Scen10	29	0.08
Scen5	30	0.083	Scen11	32	0.089
Scen6	24	0.066	Scen12	35	0.095

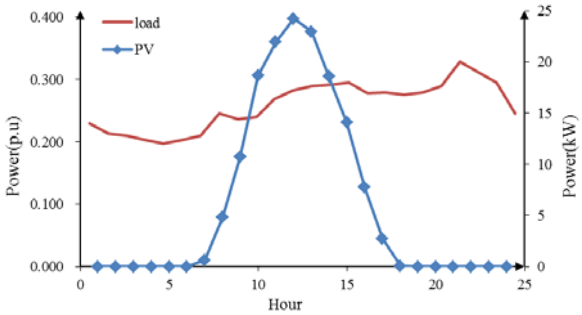

Figure 3 PV generation curving and load profile for deterministic optimization

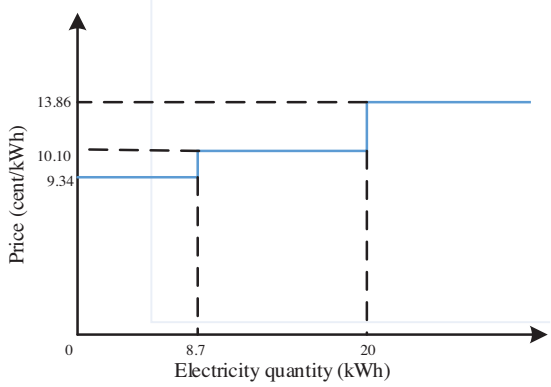

Figure 3 Prices curves under SPT

The subsidy for PV generation is 8.59 cents/kWh (0.57 ¥/kWh) and the allowance for installed capacity of PV is 30.14 $/kW (200 ¥/kW). The initial investment and operation cost of PV and BESS are taken from [8] and presented in Table 2. Social discount rate is 0.08 and the life time of PV and BESS is considered as 12 and 8 years respectively. The charging efficiency and the discharging efficiency of BESS is 0.89 and 0.91 respectively, taken from [20].

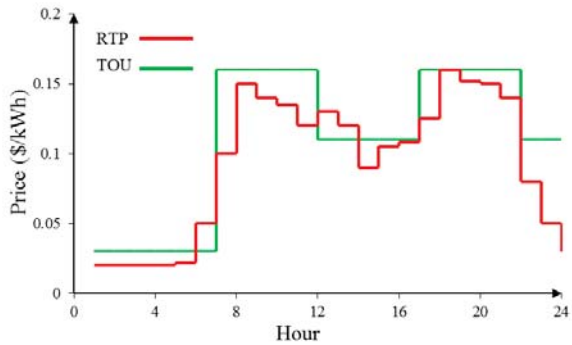

Figure 4 Prices curves under SPT

TABLE II
INITIAL INVESTMENT AND OPERATIONAL COSTS FOR BESS AND PV

Parameter	Value
PV investment cost on power	1000 ($/kW)
BESS investment cost on power	100 ($/kW)
BESS investment cost on capacity	250 ($/kWh)
PV operational cost	0.05($/kW)
BESS operational cost	0.005($/kWh)

C. Result Analysis

As shown in Table 3, the optimal results are different under the 3 tariffs. The optimal capacity of the PV system under the SPT is same as that under the TOU tariff, while it is smallest under the RTP tariff. The optimal capacity of the BESS is highest under the TOU tariff, while the RTP tariff is second and the SPT is the lowest. The reason is that the SPT provides different price gradients based on electricity consumption without the peak-valley electricity price. In this case, the benefit is lower than the investment for the BESS. Hence, the optimal capacity of the BESS is 0 under the SPT. However, there is a big peak-valley electricity price in the RTP and TOU tariffs. The BESS can be used to charge power during low-cost hours and discharge power during high-cost hours to decrease the electricity bill under the RTP and TOU. The last time of the high-cost period and low-cost period in the TOU tariff is more than that in the RTP, so that the benefit of a BESS in the TOU is greater than that in the RTP tariff. Hence, the optimal capacity of the BESS in TOU is largest.

TABLE III
OPTIMAL SIZING RESULTS OF THE PV AND BESS FOR 4 CASES

Parameters	Case1	Case2	Case3	Case4
Optimal capacity of PV (kW)	55	45	55	45
Optimal capacity of BESS (kWh)	0	35	75	0
Optimal power of BESS (kW)	0	5	10	0

Besides, the optimal results are different between SA and DA, even considering that case1 and case 4 have the same

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electricity tariff from the Table 3. The reason for this is elaborated as follows based on Table 4.

According to the decomposition of annual comprehensive costs in Table 4, the annual electricity bill in the SA optimization is nearly 29.9% lower than that in the DA optimization. In addition, the annual income from selling energy and the PV subsidy in the SA optimization is higher than that in the DA optimization. The cost-savings mentioned above are more than the increased initial investment and operation cost for the PV system in the SA optimization. Hence, the annual comprehensive cost of a home for the SA is lower than that for the DA after optimization. The annual comprehensive cost for a home in the DA is higher than that in the SA before optimization. Thus, the total cost-saving in the SA optimization is higher than that in DA optimization.

TABLE IV
THE DECOMPOSITION OF ANNUAL COST

Parameter	Case1	Case2	Case3	Case4
Annual electricity bill($/year)	12414.0	7323.2	7884.5	17713.9
Annual PV investment cost($/y)	7078.2	5791.3	7078.2	5791.3
Annual ESS investment cost($/y)	0.0	826.6	1740.2	0
Annual ESS and PV operation cost($/y)	2797.1	2398.0	3064.3	1979.2
Annual income of selling energy($/y)	851.4	265.8	596.3	16.5
Annual income of PV subsidy($/y)	4805.4	3931.7	4805.4	3400.2
Total cost($/y)	16632.6	12141.6	14365.5	17713.9

IV. CONCLUSION

The optimization of capacity configuration and operation for PV and BESS under IBR tariffs in household is studied in this paper, taking the subsidy and seasonal PV generation and load profiles into consideration. The conclusions are elaborated as follows.

- Only the PV system is equipped in a household under the STP due to the absence of peak-valley electricity price, while the PV system and BESS are equipped together under the RTP and TOU due to the existing peak-valley electricity pricing.
- That the annual comprehensive cost-savings in the SA optimization are higher than that in the DA optimization is found in case1 and case4.

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